Modeling and Validation of the Lotka–Volterra Equations Through a Food Chain Digital Twin

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Abstract

The Lotka–Volterra equations, also known as the Predator–Prey model, are first-order differential equations used to describe the dynamics of biological systems involving predation. They were independently proposed by Alfred James Lotka and Vito Volterra. This paper provides a brief overview of these equations and their theoretical results, followed by an investigation based on numerical simulations.

Keywords: Lotka–Volterra equations, Predator–Prey model, Food chain, Digital Twin, Numerical simulation.

1 Introduction

The underlying idea of these equations is the following: prey are assumed to have access to an unlimited food supply, whereas predators rely exclusively on prey as their source of food.

Let $X(t)$ and $Y(t)$ denote the prey and predator populations, respectively, at time t.

Since the prey have an unlimited food supply, their population grows proportionally to its size in the absence of predators. This can be expressed as:

$$X'(t) = aX(t), \qquad a > 0. \tag{1}$$

Conversely, in the absence of prey, the predator population can only decrease, as it has no alternative food source. Its dynamics are therefore described by:

$$Y'(t) = -bY(t), \qquad b > 0. \tag{2}$$

When both populations coexist, their dynamics are coupled through predator-prey interactions. The rate at which these interactions occur is assumed to be proportional to the probability of an encounter between individuals of the two populations, and therefore proportional to the product of their respective population sizes, $X(t)Y(t)$.

The resulting Lotka–Volterra system is given by:

$$\boxed{\begin{cases} \dfrac{dX}{dt} = aX - cXY \\ \dfrac{dY}{dt} = -bY + dXY \end{cases}} \tag{3}$$

where $a, b, c, d > 0$ are parameters describing the population dynamics and predator-prey interactions.

2 Numerical Simulations

The following diagram illustrates the code structure used to model both a simple predator–prey system and a more complex food chain, in which one predator can also hunt another predator.

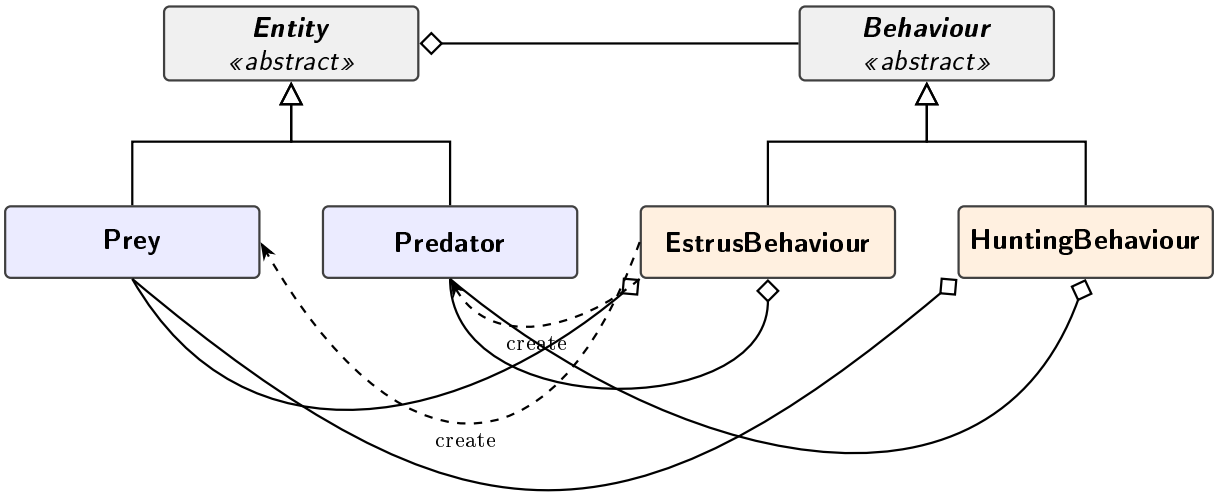


Figure 1: UML class diagram illustrating the software architecture of the food chain digital twin.

2.1 Simulation Setup

The simulation takes place on an infinite plane (or one sufficiently large to ensure that entities never reach its boundaries). Each population is assigned a movement speed, a geographical area, and a set of interaction types. Regarding hunting, we assume that when a predator chases a prey, the chase ends as soon as the predator catches it, at which point the prey is removed from the simulation.

2.2 Results

Here is an example obtained with an initial population of 150 prey and 4 predators and with $v_{predators} = 1.5 \times v_{prey}$.

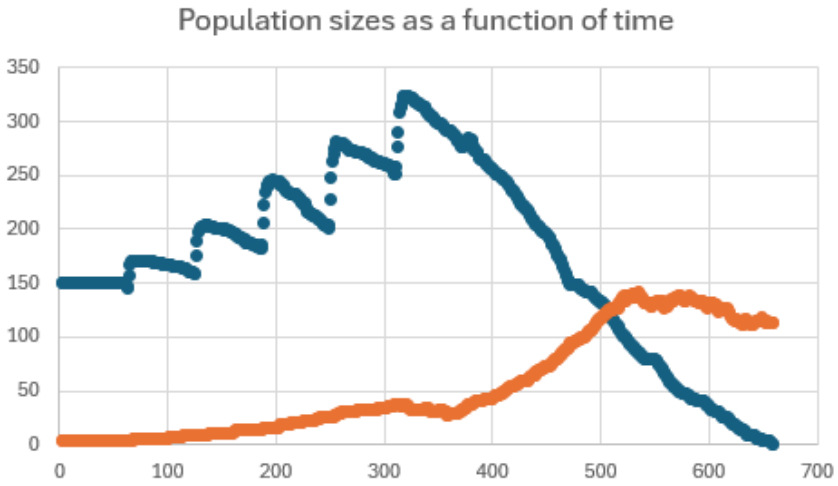


Figure 2: Simulation example - Prey and predators as a function of time

3 Calculation of constants

For a given simulation, determining the parameters of the Lotka–Volterra equations amounts to performing two linear regressions.

3.1 Linear model

We begin by reducing the problem to the case of a linear system :

$$\begin{cases} \frac{1}{X}\frac{dX}{dt} = a - cY \\ \frac{1}{Y}\frac{dY}{dt} = -b + dX \end{cases} \tag{4}$$

where $X(t), Y(t) > 0$.

Let's set,

$$\begin{cases} Z_x = \frac{1}{X}\frac{dX}{dt} = a - cY \\ Z_y = \frac{1}{Y}\frac{dY}{dt} = -b + dX \end{cases} \tag{5}$$

When comparing with a linear regression,

$$\begin{cases} Z_x = a_x + b_x Y + \epsilon_x \\ Z_y = a_y + b_y X + \epsilon_y \end{cases} \tag{6}$$

where,

$$b_x = \frac{\sum_{i=1}^{n}(Y_i - \bar{Y})(Z_{x,i} - \bar{Z}_x)}{\sum_{i=1}^{n}(Y_i - \bar{Y})^2} \tag{7}$$

$$a_x = \bar{Z}_x - b_x\bar{Y} \tag{8}$$

$$b_y = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Z_{y,i} - \bar{Z}_y)}{\sum_{i=1}^{n}(X_i - \bar{X})^2} \tag{9}$$

$$a_y = \bar{Z}_y - b_y\bar{X} \tag{10}$$

4 Example

If we return to the first implementation, shown in Figure 2, we arrive at the following analysis. A discerning eye will quickly notice an anomaly—a point that serves as a useful prompt for further reflection, and which has therefore been retained.

4.1 Linear regression

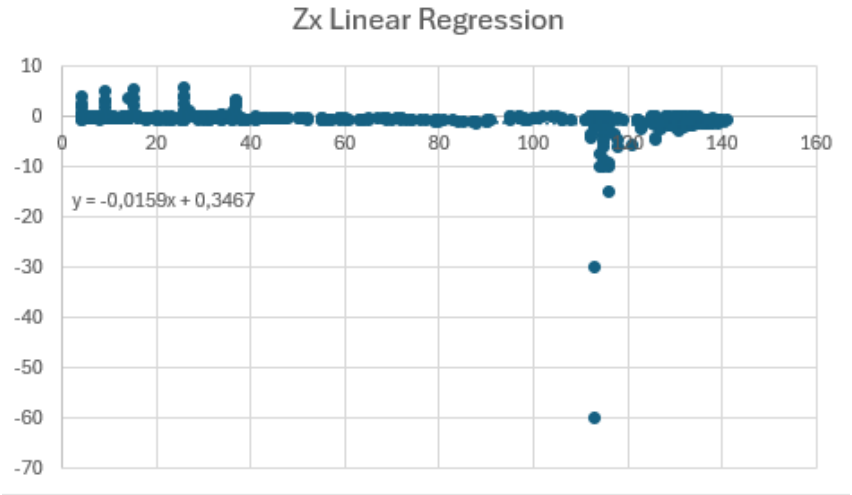


Figure 3: Simulation example - Zx linear regression

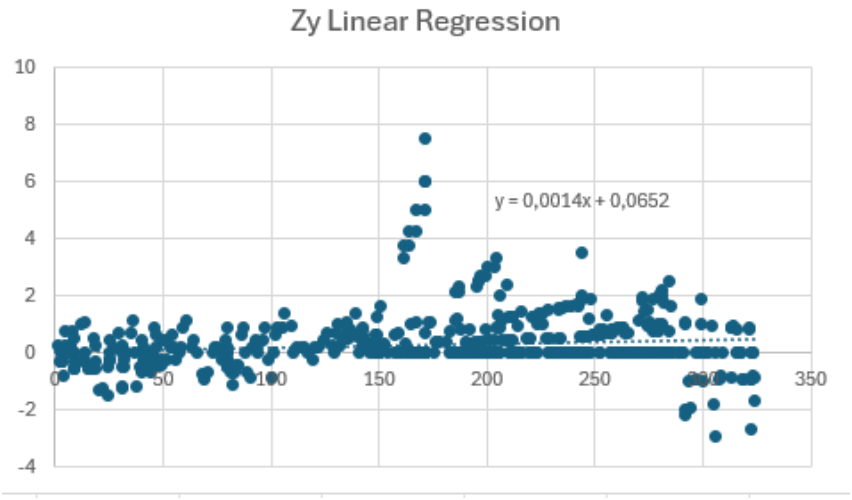


Figure 4: Simulation example - Zy linear regression

4.2 Linear system

We obtain the system of equations associated with the simulation:

$$\begin{cases} \dfrac{dX}{dt} = 0.3467X - 0.0159XY \\ \dfrac{dY}{dt} = 0.0652Y + 0.0014XY \end{cases} \tag{11}$$

It is important to note that c = -0.0652 does not satisfy our assumption that $c > 0$. This is, therefore, a case that does not strictly adhere to the Lotka-Volterra equations; this point will be explained further below.

Even when an anomaly is present, we observe that the system represents our situation quite well. Indeed, through numerical solution:

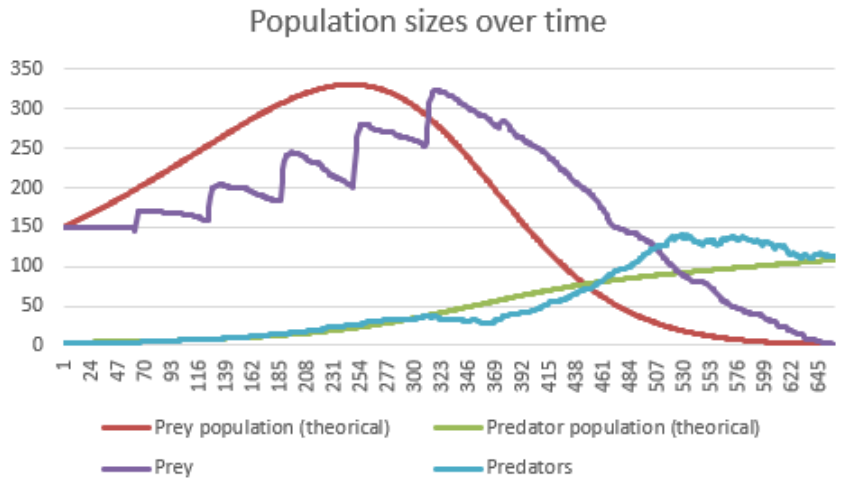


Figure 5: Simulation example - System resolution

To understand the origin of the anomaly, it should be noted that the correlation coefficients for the prey and the predators are $R^2 = 0.06545$ and $R^2 = 0.01610$, respectively—both of which

are relatively low.

This means that linear relationships do not account very well for the variations observed in the data.

It is therefore possible that the simulator does not exactly replicate the assumptions of the classic Lotka–Volterra model.

4.3 Discussion

It is noticeable that the curve representing the size of the prey population over time can be likened to a step function (at least to an extent that seems rather suspicious). The error stems from the implementation: in this version, each entity has an incremental counter—used to determine mating periods—that increases by a fixed amount each turn. Consequently, all prey enter their mating phase over the same brief time interval Δt, which creates a spurious surge. This is mathematically equivalent to multiplying the aX factor by a Dirac comb.

However, even though this explains the shape of the curve, it does not account for the relatively low correlation coefficients. The flaw here is that, in this initial implementation, a predator does not need to have consumed an excessive amount of food in order to reproduce—a factor that decouples the two variables.

4.4 Solution

Add random noise to spread out the heat periods of the entities, which will better reflect the biological reality. Make the reproduction of predators dependent on hunting—for example, by increasing hunting activity during their breeding seasons, etc.

5 Example - Hunting and reproduction combined

In this version, a slight random noise has been added to spread out the behavior of the entities over time. The predator algorithm now takes the previous remarks into account.

6 Discussion

7 Conclusion

A Additional Mathematical Details

B Additional Simulation Results